LOANZO FINANCIAL TECHNOLOGIES

Software Requirements Specification (SRS)

Peer-to-Peer Microfinance, Purpose-Bound Social Lending & Contract Synthesis Platform

Compliant with IEEE Std 830-1998 / IEEE/ISO/IEC 29148:2018 Standards

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Target Environment	Android 8.0+ (API 26-35) / Room SQLite / Cloud KMS CMEK Storage

Executive Summary: Grounded in empirical data from the **NSSO All-India Debt and Investment Survey (AIDIS, 77th Round)** and **RBI Financial Stability Reports**, informal non-institutional debt comprises 19% to 25% of household liabilities in India (where household debt has reached 45.5% of GDP). Traditional informal credit is burdened by predatory usury (24% to 120% APR) and protracted civil recovery suits (7 to 10 years). The Loanzo Protocol establishes an institutional-grade, zero-float non-custodial architecture (INR 0.00 customer deposits) operating under RBI NBFC-P2P Master Directions (RBI/2024-25/63) and RBI Penal Charges Norms (RBI/2023-24/53). It enables direct UPI rails with server-side webhook verification, mandatory digital e-stamping under Article 49 of the Indian Stamp Act 1899, Section 4 NI Act Promissory Notes enforceable via Order 37 CPC fast-track summary suits, automated milestone tranching, 100% transit fidelity collateral insurance, and multi-tenant Maker-Checker administrative governance.

Document Structure & IEEE 830 Content Map

IEEE Section	Section Title & Major Coverage Areas	Target Scope
1.0	**Introduction:** Purpose, Scope, Definitions, Empirical Sources, and Reading Guide	Foundation
2.0	**Overall Description:** Product Perspective, User Classes, Environment, and Constraints	System Context
3.0	**External Interface Requirements:** User, Hardware, Software, and Communications Interfaces	Boundary Interfaces
4.0	**System Features & Functional Requirements:** FR-01 through FR-13 Granular Engineering Specifications	Core Logic
5.0	**Non-Functional Requirements:** Performance, Security, Reliability, and Statutory Compliance	Quality & Law
6.0	**Other Architectural Requirements:** Database Entities, Room/Firestore Sync, Disaster Recovery	Data & Storage
7.0	**Verification & Traceability:** Requirements Traceability Matrix (RTM) & Test Suite Mapping	Assurance
Appendices	**Appendices A to C:** Financial Glossary, State Transitions, and Workflow Diagrams	Reference

1.0 Introduction

1.1 Purpose: This document specifies all functional and non-functional requirements for the Loanzo Peer-to-Peer Microfinance and Purpose-Bound Social Lending Platform. It serves as the primary technical specification for developers, quality assurance engineers, security auditors, and legal compliance reviewers.

1.2 Scope: Loanzo is an Android mobile platform that transforms informal lending into a secure, transparent, and court-enforceable financial architecture. The platform operates on a **Non-Custodial Zero-Float Model**, maintaining exactly INR 0.00 customer deposits. Funds route directly between verified borrower, lender, and merchant accounts via National Payments Corporation of India (NPCI) Unified Payments Interface (UPI) rails.

1.3 Definitions & Acronyms: Key terms conform to standard financial engineering practices in India: *LTV* (Loan-to-Value), *DPD* (Days Past Due), *VPA* (Virtual Payment Address), *Penny-Drop* (Automated INR 1.00 verification check), *Form 34* (Vehicle hypothecation endorsement), *Order XXXVII* (Expedited summary civil lawsuit), *Non-Compounding Late Fee* (Fair penalty calculated strictly on overdue principal), and *Digital E-Stamping* (Article 49 revenue stamp procurement).

1.4 References & Valid Data Sources: IEEE Std 830-1998; ISO/IEC/IEEE 29148:2018; NSSO MoSPI All-India Debt and Investment Survey (AIDIS, 77th Round); RBI Financial Stability Report (June 2026); RBI Master Direction NBFC-P2P (RBI/2024-25/63 DoR.FIN.REC.35/03.10.124/2024-25); RBI Fair Lending Practice - Penal Charges in Loan Accounts (RBI/2023-24/53 DoR.MCS.REC.28/01.01.001/2023-24); Indian Stamp Act 1899 (Article 49, Section 35); IT Act 2000 Sec 10A/43A/72A; NI Act 1881 Sec 4; CPC 1908 Order XXXVII; BSA 2023 Sec 63; DPDP Act 2023; IBJA / RBI Tiered Gold LTV Framework.

2.0 Overall Description

2.1 Product Perspective: Loanzo operates as an autonomous Android application combining an offline-first Room SQLite persistence layer with Cloud Firestore synchronization. It interfaces with external identity services (DigiLocker), payment networks (NPCI UPI), enterprise cloud storage (Cloud KMS CMEK), digital e-stamping gateways (NeSL), and automated messaging channels (Telegram Bot API).

2.2 User Classes and Characteristics:

* **Unified Consumer (Borrower / Lender):** End users who originate or fund micro-loans. Can toggle seamlessly between borrowing and lending perspectives.
* **Certified Field Verification Agent:** Empaneled doorstep verification officers who conduct physical collateral assaying, verify borrower residences, and transfer pledged assets into partner bank lockers.
* **System Administrators (Maker-Checker RBAC):** Multi-tenant administrative operators enforcing dual-authorization policies for reviewing agent applications, inspecting flagged KYC documents, and arbitrating multi-party grievances.

2.3 Operating Environment: Android 8.0 (API 26) through Android 15 (API 35); baseline 2 GB RAM; front-facing camera (min 2 MP); biometric sensor (Fingerprint / Face); GPS location provider with mock-location spoofing detection; network connectivity over 2G/3G/4G/5G/Wi-Fi with full offline-first degradation resilience.

2.4 Design & Regulatory Constraints: (a) Non-Custodial: Zero float balance holding; (b) Penalty Ceiling: Maximum 2.0% monthly simple non-compounding penalty per RBI/2023-24/53; (c) Mandatory Digital E-Stamping: Required under Article 49 Indian Stamp Act 1899 to cure Section 35 admissibility bar; (d) Data Localization: Local encryption using AES-256-GCM via Android StrongBox KeyStore and remote Cloud KMS CMEK under DPDP Act 2023; (e) Social Lending Cap: Aggregate INR 10 Lakh per lender ceiling.

3.0 External Interface Requirements

Interface Class	Protocol / Specification	Functional Responsibility & Operational Details
User Interface (UI)	Material Design 3 / Jetpack Compose	Single Activity architecture driving reactive declarative Compose views. Dynamic dark/light theme engine, tactile swipe-to-confirm repayment buttons, and 180-degree credit score speedometer.
Hardware Interface	BiometricPrompt, CameraX, Mock-GPS Guard	Accesses device biometric sensors for authentication, CameraX + ML Kit for liveness selfie anti-spoofing, and FusedLocationProviderClient with active mock-location spoofing detection.
Software Interface	Room SQLite, Cloud Firestore, Cloud KMS, NeSL	Room for local offline storage; Firestore for cross-device sync; Cloud Storage with CMEK encryption; NeSL / State E-Stamping API for Article 49 digital stamp certificates; Bank PA webhooks for payment reconciliation.
Communications	HTTPS TLS 1.3, FCM, SMS Retriever API	Enforces TLS 1.3 with certificate pinning; Firebase Cloud Messaging for urgent alerts; Google Play SMS Retriever API (0 SMS permission) for penny-drop matching; HTTPS Webhooks for Telegram Assistant Bot.

4.0 System Features & Functional Requirements (Part 1)

Requirements catalog with deterministic triggers, validation logic, outputs, and verifiable acceptance criteria:

Req ID	Feature Domain	Inputs, Detailed Business Logic & Acceptance Criteria	Priority
FR-AUTH-01	**Universal Identity & Session Security**	**Inputs:** Username, Email, Phone, Biometrics. **Logic:** Validates credentials across SHA-256 hashes and BiometricPrompt. Enforces banking session lifecycle: 3-min background grace window, session lock shield, 48-hr hard token purge, and device UID binding. **Acceptance Criteria:** Launch to interactive state in <= 1.8s; zero session hijacking.	Mandatory
FR-KYC-02	**DigiLocker & Encrypted Vault**	**Inputs:** Aadhaar number, PAN, camera selfie. **Logic:** Verifies authentic identity records via DigiLocker. Captures ML Kit liveness selfie. Houses credentials inside an AES-256 encrypted vault gated behind password/biometric challenge. Remote backups protected via Cloud KMS CMEK under DPDP Act. **Acceptance Criteria:** 100% of KYC credentials encrypted at rest; zero unauthenticated views.	Mandatory
FR-BANK-03	**Penny-Drop & Webhook Reconcile**	**Inputs:** Recipient Account, IFSC, server PA webhook, SMS Retriever. **Logic:** Executes automated INR 1.00 test transfer. Reconciles via server-side Payment Aggregator webhooks to prevent client-side intent spoofing. Matches recipient name against KYC record: >=95% auto-verified, 80-94% dual-admin review, <80% freeze. Max 3 retries. **Acceptance Criteria:** Capital transfer blocked on <80% match; zero dangerous SMS permissions.	Mandatory
FR-MKT-04	**Social Marketplace & Bidding**	**Inputs:** Loan requests, capital pools, counter-bids. **Logic:** Categorized public discovery feed (#Medical, #Business, etc.). Community vouches boost trust score. Lenders submit competing interest rate bids; borrowers execute 1-tap agreement finalization. **Acceptance Criteria:** Competitive bids display side-by-side; real-time Firestore sync.	High
FR-LOAN-05	**Purpose-Bound Tranche Payout**	**Inputs:** Milestone commercial invoice, seller UPI VPA. **Logic:** Verifies merchant invoice and past milestone completion. Disburses loan tranches directly to seller UPI VPA. Discrepancies halt release for admin inspection. Zero capital released as unmonitored cash. **Acceptance Criteria:** 100% of milestone disbursements settle into verified seller VPAs.	Mandatory
FR-COL-06	**Tiered LTV & Insured Custody**	**Inputs:** Asset category, weight, appraisal proof. **Logic:** Enforces statutory tiered LTV: Gold (up to 75%-85% on IBJA rates per RBI framework), Property (55%), Vehicles (60%), Equipment (45%). Doorstep agent assays net weight. Pledged transit covered by 100% fidelity insurance with Dual-OTP handover (doorstep to bank vault). **Acceptance Criteria:** Programmatically blocks loans exceeding LTV; 100% insured transit.	Mandatory

4.0 System Features & Functional Requirements (Part 2)

Req ID	Feature Domain	Inputs, Detailed Business Logic & Acceptance Criteria	Priority
FR-AGR-07	**Digital E-Stamping & Sec 4 NI Note**	**Inputs:** Digital stamp cert, signature canvas, selfie, biometric consent. **Logic:** Interfaces with NeSL/State e-stamping API to procure digital stamp certificate under Art 49 Indian Stamp Act 1899. Synthesizes Section 4 NI Promissory Note with 3-factor eSign and Section 63 BSA 2023 certificate, curing Section 35 court admissibility bar. **Acceptance Criteria:** Digital stamp duty affixed; PDF generation in <= 2.5s; SHA-256 verifiable.	Mandatory
FR-PAY-08	**Dynamic QR & Server-Reconciled Pay**	**Inputs:** Due date, installment amount, prepayment. **Logic:** Dynamic UPI QR generation (`upi://pay`). Tactile swipe-to-confirm button eliminates accidental debits. Reconciled via server-side Payment Aggregator webhooks before updating loan ledger. Prepayment simulator computes interest saved. **Acceptance Criteria:** Zero client-side intent spoofing; zero accidental double debits.	Mandatory
FR-DEL-09	**Non-Compounding & Order 37 Dossier**	**Inputs:** Overdue days count (DPD). **Logic:** 3-day grace window (INR 0.00 fee). From Day 4, simple daily fee capped at 2.0%/month (RBI/2023-24/53). At DPD > 30, calls guarantor. At DPD > 90, compiles Order 37 CPC summary suit dossier (e-stamped note, BSA 2023 cert) eliminating oral evidence stages. **Acceptance Criteria:** Late fee never capitalized into principal; Order 37 dossier generated.	Mandatory
FR-AGT-10	**Field Agent Empanelment & Dispatch**	**Inputs:** Verification experience, Police Clearance (PCC). **Logic:** Bank-grade empanelment reviewed via Maker-Checker console. Strict role isolation permanently hides consumer borrowing. Geotagged inspections with mock-GPS detection and instant INR 500 to INR 1,500 per-visit compensation crediting. **Acceptance Criteria:** Unapproved agents blocked from console; 100% geotagged inspections.	High
FR-ADM-11	**Multi-Tenant Maker-Checker Desk**	**Inputs:** Administrator RBAC credentials. **Logic:** Multi-tenant admin console with mandatory Maker-Checker dual authorization for agent approvals, KYC overrides, and dispute escrow release, eliminating single points of failure. Virtual dispute mediation calendar with Google Meet links. **Acceptance Criteria:** Dual-authorization enforced on high-risk actions; real-time dispatch queue.	High
FR-COM-12	**Multilingual & Telegram Assistant**	**Inputs:** Chat messages, bot commands, language choice. **Logic:** High-speed LRU translation across 21+ languages (<150ms). 24/7 Telegram Assistant Bot (@Loanzo_bot) dispatches automated EMI reminders and administrative alerts with cryptographic token binding. **Acceptance Criteria:** Multi-language UI switching in real-time; 100% bot webhook reliability.	High
FR-AUD-13	**Credit Health & Legal NOC Settlement**	**Inputs:** Repayment history, zero-due balance. **Logic:** 180-degree speedometer credit gauge (score 0 to 100). Immutable SHA-256 audit trail. Upon INR 0.00 balance, issues official No Objection Certificate (NOC) with digital seal, de-hypothecating collateral within 24 hours. **Acceptance Criteria:** Official NOC PDF generated with cryptographic verification seal.	Mandatory

5.0 Non-Functional Requirements

Req ID	Quality Attribute	Operational Requirement & Quantitative Threshold	Verification Standard
NFR-PERF-01	**Database Latency**	Local Room SQLite queries execute within <= 15 ms (95th percentile) on baseline 2 GB RAM Android hardware.	Macrobenchmark Suite
NFR-PERF-02	**Server Concurrency**	Cloud backend sustains >= 2,500 requests/sec during peak hours with 95th percentile latency < 250 ms.	Distributed Load Test
NFR-SEC-01	**Encryption at Rest**	Local records encrypted via AES-256-GCM StrongBox; remote storage protected with Cloud KMS CMEK under DPDP Act.	Key Management Audit
NFR-SEC-02	**Encryption in Transit**	All network communications enforce TLS 1.3 with SHA-256 certificate pinning across all REST endpoints.	Traffic Interception Test
NFR-SAFE-01	**Non-Custodial Float**	Platform shall maintain exactly INR 0.00 custodial float balance. Zero pooling of funds permitted.	Ledger Balance Audit
NFR-SAFE-02	**Transit Fidelity Cover**	100% of pledged physical collateral in transit between doorstep and bank vault is covered under fidelity insurance.	Policy Underwriting Audit
NFR-REG-01	**RBI Penalty Compliance**	Late charges accrue strictly as simple flat fee capped at 2.0%/month; zero capitalization into principal.	RBI/2023-24/53 Audit
NFR-REG-02	**E-Stamping & BSA 2023**	Digital e-stamping under Art 49 Stamp Act 1899 and electronic certificates under Sec 63 BSA 2023.	Judicial Admissibility Audit

6.0 Database Requirements & Operational Architecture

Entity Name	Primary Key & Key Fields	Relational Role & Operational Purpose
UserEntity	userId, phone, username, kycStatus, registeredDeviceId, telegramChatId	Stores core profile, role flags (CONSUMER vs AGENT), and biometric registration status.
LoanEntity	loanId, lenderId, borrowerId, guarantorId, principal, interestRate, status	Tracks the lifecycle of each loan from DRAFT, ESIGNING, ACTIVE, DELINQUENT, to SETTLED.
RepaymentEntity	repaymentId, loanId, installmentNumber, dueDate, paidDate, amount, serverWebhookId	Maintains EMI schedule, tracking paid installments, due dates, simple late fees, and server webhook IDs.
PledgeEntity	pledgeId, loanId, assetClass, grossWeight, netWeight, insurancePolicyNumber, lockerNumber	Records physical collateral details, doorstep assaying results, fidelity insurance policy, and bank locker.
AgreementEntity	agreementId, loanId, agreementPdfUrl, eStampCertificateNumber, contractSha256, gpsCoordinates	Houses digital Promissory Note PDF links, NeSL e-stamp certificate numbers, SHA-256 hashes, and execution geotags.
AgentInspectionEntity	visitId, agentId, loanId, visitType, checkInGps, mockGpsFlag, compensationAmount	Tracks doorstep verification assignments, GPS check-in proof, mock-GPS flags, and per-visit agent payouts.
NocCertificateEntity	nocId, loanId, zeroDueTimestamp, digitalSignatureHash, certificatePdfPath	Stores legally binding NOC certificates issued upon INR 0.00 balance, triggering collateral release within 24 hours.

7.0 Requirements Traceability Matrix (RTM)

Req ID	Requirement Summary	Primary Design Module	Automated Test ID	Statutory Basis
FR-AUTH-01	Universal Identifier Login	**AuthViewModel.kt**	**TC-AUTH-101**	IT Act Sec. 43A
FR-AUTH-04	Banking Session Lifecycle	**SessionManager.kt**	**TC-AUTH-104**	RBI IT Framework
FR-KYC-02	Liveness & Encrypted Vault	**DocumentVaultScreen.kt**	**TC-KYC-203**	DPDP Act Sec. 8
FR-BANK-03	Penny-Drop & Webhook Match	**BankVerificationService.kt**	**TC-BNK-302**	RBI P2P Directions
FR-MKT-04	Social Competitive Bidding	**MarketplaceViewModel.kt**	**TC-MKT-401**	Fair Practices Code
FR-LOAN-05	Milestone Tranche Disbursement	**TrancheDisbursalEngine.kt**	**TC-LON-501**	P2P Guidelines
FR-COL-06	Tiered LTV & Insured Custody	**CollateralValuationEngine.kt**	**TC-COL-604**	Contract Act Sec. 176
FR-AGR-07	Digital E-Stamping & Promissory	**AgreementGenerator.kt**	**TC-AGR-702**	Stamp Act Art. 49
FR-PAY-08	Server Webhook QR Repayment	**PaymentRepository.kt**	**TC-PAY-802**	Payment Systems Act
FR-DEL-09	Non-Compounding & Order 37	**PenaltyCalculator.kt**	**TC-DEL-902**	RBI/2023-24/53 / CPC
FR-AGT-10	Field Agent Empanelment & PCC	**AgentApplicationScreen.kt**	**TC-AGT-1001**	Police Norms
FR-ADM-11	Multi-Tenant Maker-Checker Desk	**AppOwnerVerificationScreen.kt**	**TC-ADM-1101**	Corporate Governance
FR-AUD-13	Official NOC & Credit Gauge	**NocGenerator.kt**	**TC-AUD-1301**	IT Act 2000 Sec. 10A

Appendix A: Plain-English Glossary of Starred & Financial Terms

Term	Plain-English Definition & Real-World Meaning
Tranche Disbursement	Releasing a loan in smaller milestone-bound installments (e.g. tools first, materials second) tied to verified seller bills rather than handing over unmonitored cash upfront.
Penny-Drop Check	Sending a tiny test payment of INR 1.00 to an account to automatically fetch and verify the recipient's real registered bank name before releasing loan capital.
Digital E-Stamping	Procuring official state revenue stamp certificates electronically (under Article 49 of the Indian Stamp Act 1899) to satisfy Section 35 and ensure court admissibility.
Maker-Checker Governance	Enterprise dual-authorization rule requiring one administrator to propose and a second independent administrator to approve high-risk actions, preventing single points of failure.
UPI VPA	Virtual Payment Address (e.g. dealer@bank) -- a unique UPI identifier used to route digital money transfers directly to verified merchants without exposing raw bank details.
Tiered LTV Ratio (IBJA)	Maximum loan percentage allowed against gold purity closing prices published by IBJA (up to 85% for loans <= INR 2.5L, 80% <= INR 5L, and 75% > INR 5L per RBI framework).
Form 34 Hypothecation	A formal legal endorsement recorded on a vehicle's official registration certificate (RC Book) noting that the vehicle acts as collateral for the loan.
Promissory Note (Sec 4 NI Act)	A legally binding document signed by the borrower containing an unconditional written promise to repay a specific sum of money to the lender on a specified date.
Simple Late Fee (Non-Compounding)	A flat daily fee on overdue installments that can never be added into loan principal, preventing lenders from charging 'interest-on-interest' (debt traps).
Order XXXVII (Order 37 CPC)	A special fast-track civil court lawsuit in India for debt recovery based on written promissory notes, resolving cases without long 7-year trials.
Transit Fidelity Insurance	An institutional insurance policy underwriting 100% of pledged physical collateral value during agent transit from the borrower's doorstep to the partner bank locker.